Structural Free Energy and the Critical Coherence Index in a Continuous Ensemble of ϕ^4 Field Configurations

An Extended Numerical Study with Mixed Initial Conditions

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Abstract

The companion paper [5] tested the hypothesis that the Critical Coherence Index (CCI) correlates with the ratio of coarse-grained histogram activity to structural mutual information, using a discrete ensemble of four configuration classes. A limitation of that study was that CCI values clustered near zero for all but the chaotic class, making the correlation dependent on a single class separation.

The present paper addresses this by employing a *continuous mixed ensemble*: $N = 160$ trajectories whose initial conditions are random linear combinations of vacuum, kink, localised, and noise components, producing a broad and continuous distribution of dynamical states.

The key findings are:

1. The CCI correlates *strongly* with the structural free energy $\mathcal{F}_{\text{struct}}$ across the continuous ensemble (Spearman $r_s = 0.878$, $p < 10^{-50}$).
2. The CCI correlates *weakly* with the coarse-grained histogram activity to mutual information ratio $\dot{S}_{\text{cg}}/I_{\text{nn}}$ (Spearman $r_s = 0.288$, $p = 2 \times 10^{-4}$).
3. The CCI correlates *negatively* with mutual information alone ($r_s = -0.341$), indicating that high structural information suppresses the CCI.

These results refine the interpretation of the CCI: it is primarily a measure of *structural free energy*, not of entropy-to-information ratio. The connection to histogram activity and mutual information is real but secondary. The structural free energy $\mathcal{F}_{\text{struct}}$ emerges as the more fundamental quantity.

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1 Introduction

The Critical Coherence Index (CCI), introduced in [4], is a dimensionless ratio designed to classify field configurations into ordered, critical, and chaotic dynamical regimes. The companion paper [5] proposed and partially confirmed the hypothesis that the CCI approximates the ratio between coarse-grained histogram activity and structural mutual information.

However, that study used a discrete ensemble of four rigid configuration classes (vacuum, kink, localised, chaotic), which led to a clustering of CCI values near zero for three of the four classes. The reported Spearman correlation of $r_s = 0.76$ was therefore driven primarily by the separation of the chaotic class from the others — a limitation acknowledged in [5].

The present paper addresses this limitation by replacing the discrete class ensemble with a *continuous mixed ensemble*: $N_{\text{runs}} = 160$ trajectories with initial conditions that are random linear combinations of four component types (vacuum, kink, localised, noise), sampled continuously from the unit hypercube of mixing coefficients. This produces a broad, continuous distribution of CCI values and structural properties, enabling a more stringent test of the proposed correlations.

2 Model and Ensemble

2.1 Field Theory

We study the same ϕ^4 scalar field as in the companion papers [2, 5]:

$$\ddot{\phi}_i = \frac{\phi_{i+1} + \phi_{i-1} - 2\phi_i}{\Delta x^2} - V'(\phi_i), \qquad V(\phi) = \tfrac{1}{4}(\phi^2 - 1)^2, \tag{1}$$

on a periodic lattice of $N = 128$ sites ($\Delta x = 1$), integrated with fourth-order Runge–Kutta ($\Delta t = 0.02$, $T_{\max} = 24$, seed $= 42$).

2.2 Continuous Mixed Ensemble

Rather than assigning each trajectory to a fixed configuration class, we sample initial conditions from a continuous mixture:

$$\phi_0(x) = \frac{a_{\text{vac}}\,\phi_{\text{vac}}(x) + a_{\text{kink}}\,\phi_{\text{kink}}(x) + a_{\text{loc}}\,\phi_{\text{loc}}(x) + a_{\text{noise}}\,\phi_{\text{noise}}(x)}{1 + \sigma[\phi_0]}, \tag{2}$$

where $a_i \sim \text{Uniform}(0, 1)$ independently, $\sigma[\phi_0]$ is the standard deviation of the unnormalised sum (a gentle normalisation to avoid extreme amplitudes), and the field is clipped to $[-2.5, 2.5]$.

This produces $N_{\text{runs}} = 160$ trajectories spanning a continuous range of vacuum-like, kink-like, localised, and chaotic-like states, without hard class boundaries. The ensemble size of 160 is sufficiently large for stable rank correlations: bootstrapped Spearman estimates show standard errors below 0.05 for all reported correlations.

2.3 Measured Quantities

For each trajectory we record time-averaged values of:

1. $\overline{\mathcal{C}}_{\text{CCI}}$ — mean CCI along the trajectory.
2. $\mathcal{F}_{\text{struct}}$ — structural free energy [4].

3. $\overline{\dot{S}}_{\mathrm{cg}}$ — mean rate of change of coarse-grained histogram entropy (32 bins), interpreted as a proxy for dynamical activity (not thermodynamic entropy production; the system is Hamiltonian and the true phase-space entropy is conserved by the Liouville theorem).

4. $\overline{I}_{\mathrm{nn}}$ — mean nearest-neighbour mutual information (16 bins).

5. $\overline{\dot{S}}_{\mathrm{cg}}/(\overline{I}_{\mathrm{nn}} + \varepsilon)$ — the ratio proposed in [5].

3 Results

3.1 Correlation Summary

Table 1 reports the Spearman and Pearson correlations between the mean CCI and the candidate quantities across the 160-trajectory ensemble.

Table 1: Spearman and Pearson correlations with mean CCI ($N = 160$ continuous-mixture trajectories).

Quantity	Spearman r_s	p-value	Pearson r	p-value
$\mathcal{F}_{\mathrm{struct}}$	**0.878**	$< 10^{-50}$	**0.885**	$< 10^{-50}$
$\overline{\dot{S}}_{\mathrm{cg}}/(\overline{I}_{\mathrm{nn}} + \varepsilon)$	0.288	2×10^{-4}	-0.015	0.85
$\overline{\dot{S}}_{\mathrm{cg}}$ alone	-0.131	0.098	-0.037	0.64
$\overline{I}_{\mathrm{nn}}$ alone	-0.341	$< 10^{-5}$	-0.338	$< 10^{-5}$

Three findings stand out:

Finding 1. The structural free energy $\mathcal{F}_{\mathrm{struct}}$ correlates with the CCI at $r_s = 0.878$ ($p < 10^{-50}$), a highly significant and robust result across the continuous ensemble. This is the strongest correlation observed and confirms that the CCI is fundamentally a measure of structural free energy.

Finding 2. The ratio $\dot{S}_{\mathrm{cg}}/I_{\mathrm{nn}}$, proposed in [5] as a physical proxy for the CCI, shows only a moderate Spearman correlation ($r_s = 0.288$) and a near-zero Pearson correlation ($r = -0.015$). The Spearman–Pearson discrepancy suggests a monotonic but highly nonlinear relationship. The hypothesis of [5] is therefore only partially supported in the continuous ensemble.

Finding 3. Mutual information I_{nn} correlates *negatively* with the CCI ($r_s = -0.341$). This is consistent with the design of the CCI: high spatial correlations (large I_{nn}) appear in the denominator and suppress the index, so configurations with rich structural information have lower CCI values.

3.2 MaxEnt Selection

Using the structural free energy to define an effective Boltzmann weighting (not a thermodynamic temperature, but a selection parameter β controlling preference sharpness):

$$P[\Phi] \propto \exp(-\beta\, \mathcal{F}_{\mathrm{struct}}[\Phi]), \quad \beta = 4,$$

the top-weighted trajectories are dominated by configurations with large kink and localised components ($a_{\mathrm{kink}} \approx 0.5$–$1.0$, $a_{\mathrm{loc}} \approx 0.25$–$0.95$) and very small noise ($a_{\mathrm{noise}} < 0.16$). This is consistent with the 1D kink-dominance result of [2]: the structural selection principle prefers topologically structured, low-activity configurations.

4 Discussion

4.1 CCI as Structural Free Energy Proxy

The dominant finding of this study is that the CCI is primarily a proxy for the structural free energy $\mathcal{F}_{\text{struct}}$, not directly for the histogram-activity-to-information ratio. This is consistent with the construction of the CCI in [4]: the numerator (instability $\times$ activity $\times$ imbalance) and denominator (sum of coherence diagnostics) together provide a dimensionless proxy for the structural free energy.

The strong correlation ($r_s = 0.878$) across a continuous, 160-trajectory ensemble is more robust evidence than the discrete-class result of [5], where the correlation was driven by a single class separation.

4.2 Partial Support for the Entropy-Information Hypothesis

The ratio $\dot{S}_{\text{cg}}/I_{\text{nn}}$ remains a meaningful quantity: its Spearman correlation with the CCI is statistically significant ($p = 2 \times 10^{-4}$), and the negative correlation of I_{nn} alone confirms that structural information is captured by the CCI denominator.

However, the near-zero Pearson correlation ($r = -0.015$) reveals that the relationship is strongly nonlinear. The histogram activity $\dot{S}_{\text{cg}}$ alone shows no significant correlation with the CCI ($r_s = -0.131$, $p = 0.098$), indicating that it is not a dominant driver of the index in the continuous ensemble.

A more refined physical proxy for the CCI may require a better estimator of dynamical activity — for example, Lyapunov exponents, transfer entropy, or spectral mode occupation — rather than the coarse histogram turnover rate.

4.3 Limitations

1. **Single model.** Results are for a 1D ϕ^4 field only. Extension to 2D, sine-Gordon, and other models is required.

2. **Histogram activity proxy.** As in [5], $\dot{S}_{\text{cg}}$ is a coarse proxy for dynamical activity. The system is Hamiltonian; true thermodynamic entropy production is zero by the Liouville theorem.

3. **Conservation diagnostic placeholder.** The diagnostic Γ_{cons} is set to a neutral constant in the present implementation. A proper implementation based on energy-momentum flux defects may alter the quantitative results.

5 Conclusion

We have extended the numerical test of the Critical Coherence Index to a continuous mixed ensemble of $N = 160$ field trajectories, addressing the class-clustering limitation of the companion paper [5].

The main finding is that the CCI is primarily a proxy for the structural free energy $\mathcal{F}_{\text{struct}}$ ($r_s = 0.878$, $p < 10^{-50}$), rather than directly for the histogram-activity-to-information ratio. The entropy-information hypothesis of [5] receives partial support: mutual information suppresses the CCI as designed, but the histogram activity proxy is not a strong predictor in the continuous ensemble.

These findings clarify the role of the CCI within the structural selection framework: it measures the structural free energy of a configuration, and the structural free energy in turn determines the Boltzmann selection weight. The MaxEnt selection consistently favours low-noise,

topologically structured configurations (kink-dominated trajectories), consistent with the robust kink-dominance result of [2].

Future work will investigate improved dynamical activity estimators, extension to 2D and 3D field theories, and the quantitative calibration of the CCI thresholds τ_1 and τ_2 from the continuous ensemble.

A Reproducibility

All results are reproducible from the companion code:

https://github.com/Chris4081/structural-selection-principle

Run:

```
pip install numpy pandas matplotlib scipy
python cci_struct_test.py
```

Key parameters: $N = 128$, $\Delta t = 0.02$, $T_{\max} = 24$, $N_{\text{runs}} = 160$, seed $= 42$, $N_{\text{bins,entropy}} = 32$, $N_{\text{bins,MI}} = 16$, $\beta = 4$, $\kappa = 1$, $\varepsilon = 10^{-8}$.

References

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